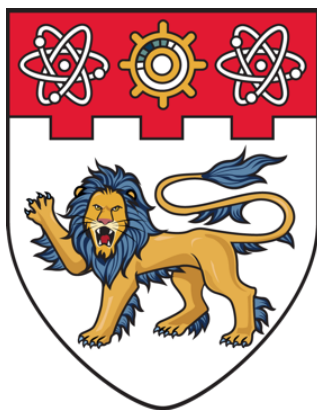




**NANYANG
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AI6126 ADVANCED COMPUTER VISION

**PROJECT REPORT ON:
I2I TRANSLATION AND INPUT SPACE UDA**

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1 Introduction

This project investigates image-to-image (I2I) translation and its application to unsupervised domain adaptation (UDA) in the input space. The central motivation is that domain shift often manifests as differences in appearance statistics, such as illumination, texture, color distribution, or sketch-like abstraction, which can significantly degrade the generalization ability of models trained only on the source domain. Instead of directly aligning latent features, this project focuses on reducing the source-target discrepancy at the image level, such that labeled source data can be transformed into target-style samples and subsequently reused for downstream supervised learning.

The study is organized into two complementary tasks. Task I compares spatial-domain and spectral-domain style transfer by benchmarking the standard CycleGAN [5] and our Spectral CycleGAN variant. The goal of this part is to analyze how different forms of style supervision affect the visual quality and domain characteristics of translated images. Building upon these observations, Task II evaluates several representative input-space UDA strategies, including Source-only, FDA, CycleGAN-based translation, Spectral CycleGAN-based translation, and CyCADA [2], in order to assess how effectively different translation mechanisms improve target-domain recognition performance.

Experiments are conducted on four cross-domain benchmarks spanning both digit recognition and object classification: MNIST $\rightarrow$ USPS, Amazon $\rightarrow$ Webcam (Office-31), Art $\rightarrow$ Real-World (Office-Home), and Photo $\rightarrow$ Sketch (PACS). These datasets exhibit diverse types of domain gaps, ranging from relatively simple appearance changes to more severe structural and semantic discrepancies. Therefore, they provide a suitable testbed for examining whether style transfer in either spatial or spectral form is sufficient for domain adaptation, and under what conditions additional semantic or feature-level alignment becomes necessary.

2 Task I: Spatial vs. Spectral Style Transfer

2.1 Methodology

2.1.1 Standard CycleGAN

CycleGAN[5] is an effective approach for Unsupervised Domain Adaptation. Its core appeal lies in breaking the constraint of traditional style transfer methods that require paired data, thereby enabling UDA without the need for strict semantic or content alignment between the two domains. According to Zhu et al. [5], while adversarial training can align the distributions of two domains, it lacks constraints to ensure that an individual input x is mapped to a specific desired output y in a meaningful way. This often leads to the 'mode collapse' or random permutation mapping. By introducing Cycle Consistency, the network is able to learn a more deterministic and structure-preserving mapping, as the model must retain enough information to reconstruct the original input from the translated output.

In addition to the cycle-consistency loss, the original work also introduces an identity loss[5], as formulated in Equation (4), which helps preserve the color profile and prevents undesirable color shifts during the style transfer process. The final loss of CycleGAN can be defined by Equations (1) through (4).

$$L_{total} = \lambda_{GAN}(L_{GAN}(G, D_Y, X, Y) + L_{GAN}(F, D_X, Y, X)) + \lambda_{cyc}L_{cyc}(G, F) + \lambda_{id}L_{identity}(G, F) \quad (1)$$

$$L_{GAN}(G, D_Y, X, Y) = \mathbb{E}_{y \sim p_{data}(y)}[\log D_Y(y)] + \mathbb{E}_{x \sim p_{data}(x)}[\log(1 - D_Y(G(x)))] \quad (2)$$

$$L_{cyc}(G, F) = \mathbb{E}_{x \sim p_{data}(x)}[\|F(G(x)) - x\|_1] + \mathbb{E}_{y \sim p_{data}(y)}[\|G(F(y)) - y\|_1] \quad (3)$$

$$L_{identity}(G, F) = \mathbb{E}_{y \sim p_{data}(y)}[\|G(y) - y\|_1] + \mathbb{E}_{x \sim p_{data}(x)}[\|F(x) - x\|_1] \quad (4)$$

Where G and F are the generators responsible for the mappings $X \rightarrow Y$ and $Y \rightarrow X$ respectively, D_Y and D_X are the adversarial discriminators for each domain, x and y are image samples from the source and

target domains, $p_{data}(x)$ and $p_{data}(y)$ denote the respective data distributions, $\|\cdot\|_1$ is the L_1 norm measuring pixel-wise absolute error, and λ_{GAN} , λ_{cyc} , and λ_{id} are the weights assigned to each loss component.

2.1.2 Spectral CycleGAN

Since standard CycleGAN operates within the spatial domain, pixel-level discriminators are often susceptible to local noise, making it difficult to effectively capture global styles such as lighting and background tones. To address this limitation, we shift our focus to the frequency domain, where global stylistic patterns are more prominently represented. However, rather than directly processing FFT-transformed images—which often results in the loss of spatial locality and increased sensitivity to phase distortions—we incorporate spectral supervision as a regularizer. Following the approach of Pai et al.[3]. The spectral loss can be defined by Equations (5) to (8).

$$L_{GSL}(G, F) = \mathbb{E}_{x \sim p_{data}(x)}[L_{gsl}(G(x), y)] + \mathbb{E}_{y \sim p_{data}(y)}[L_{gsl}(F(y), x)] \quad (5)$$

$$L_{gsl}(\hat{I}, I) = \frac{1}{M \times N} \sum_{u,v} w(u, v) \cdot \left| \log(|\mathcal{F}shift(\text{DFT}(\hat{I}))|_{u,v+1}) - \log(|\mathcal{F}shift(\text{DFT}(I))|_{u,v+1}) \right| \quad (6)$$

$$w(u, v) = \alpha \cdot (\text{Dist}_{u,v} + \epsilon)^\beta \quad (7)$$

$$\text{Dist}_{u,v} = \sqrt{\left(\frac{2u}{M-1} - 1\right)^2 + \left(\frac{2v}{N-1} - 1\right)^2} \quad (8)$$

Where x and y denote image samples from the source and target domains respectively, M and N represent the image dimensions, $\text{DFT}(\cdot)$ represents the Discrete Fourier Transform, and $\mathcal{F}shift$ indicates the centering of the zero-frequency component. The term $w(u, v)$ is a distance-based soft mask defined by the hyperparameters α and β . In our configuration, a negative value of β is employed to prioritize the alignment of low-frequency components.

This spectral loss is multiplied by a hyperparameter weight and added to the total CycleGAN loss, with the final objective defined as shown in Equation (9).

$$L_{total} = \lambda_{GAN}(L_{GAN}(G, D_Y, X, Y) + L_{GAN}(F, D_X, Y, X)) + \lambda_{cyc}L_{cyc}(G, F) + \lambda_{id}L_{identity}(G, F) + \lambda_{gsl}L_{GSL}(G, F) \quad (9)$$

2.2 Experimental Setup

We conducted image-to-image translation experiments using the two aforementioned models across three benchmarks: Amazon $\rightarrow$ Webcam (Office-31), Art $\rightarrow$ Real-World (Office-Home), and Photo $\rightarrow$ Sketch (PACS).

The model was trained with the following hyperparameter configurations for the total loss: $\lambda_{GAN} = 1$, $\lambda_{cyc} = 10.0$, $\lambda_{gsl} = 1$, and $\lambda_{id} = 0.2$. These coefficients ensure that the cycle-consistency and spectral constraints are properly weighted against the adversarial and identity objectives.

For computational efficiency, input images were first resized to a minimum side length of 128 pixels and subsequently randomly cropped to a fixed resolution of 128×128 . Furthermore, to maintain the equilibrium of the adversarial training process, the learning rate of the discriminators (D) was suppressed relative to that of the generators (G). Specifically, for the Amazon $\rightarrow$ Webcam (Office-31) and Art $\rightarrow$ Real-World (Office-Home) tasks, the learning rate for D was set to half of that for G . For the Photo $\rightarrow$ Sketch (PACS) task, the discriminator’s learning rate was further reduced to one-fifth of the generator’s.

All models were trained for 200 epochs using a cosine annealing scheduler with an initial learning rate of 0.0005.

2.3 Results

Figure 1 showcases selected examples of the style transfer outcomes across different datasets. Observing the translation outcomes, both models yield intuitive style transfer effects on the Amazon $\rightarrow$ Webcam and Photo $\rightarrow$ Sketch benchmarks. However, the transformation on the Art $\rightarrow$ Real-World dataset appears less pronounced. Furthermore, the reconstructed images from both models exhibit a certain degree of blurring, loss of fine details, and ripple-like artifacts.

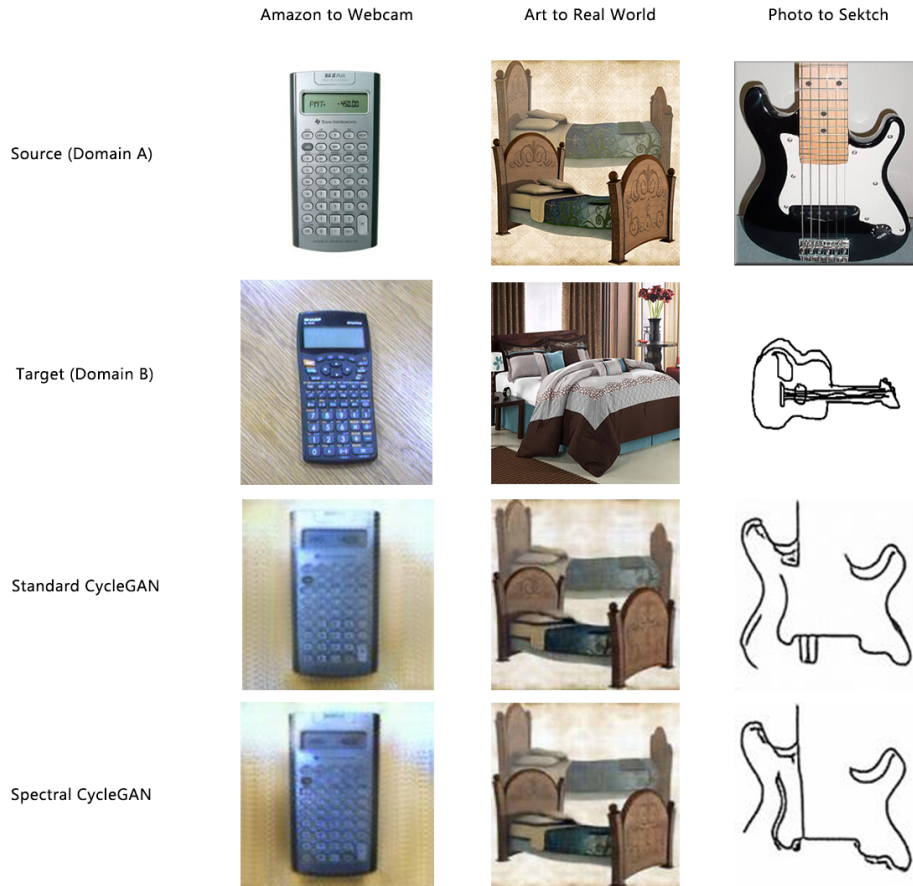


Figure 1: UDA Samples

However, it is challenging to discern the superiority of these two methods through visual inspection alone. Therefore, it is necessary to introduce downstream tasks for a quantitative comparison. This will be discussed in detail in the following section, 'Task II: Benchmarking UDA Approaches'.

3 Task II: Benchmarking UDA Approaches

3.1 Compared Methods

3.1.1 Source-only

We selected ResNet-18 [1] as the classifier for the downstream task. In this approach, the pre-trained ResNet-18 model is fine-tuned directly using images from the source domain, followed by an evaluation of its classification accuracy on the target domain.

3.1.2 FDA

Fourier Domain Adaptation (FDA) performs pixel-level domain adaptation in the frequency domain by replacing the low-frequency amplitude component of a source image with that of a target-domain image, while preserving the phase information of the source image. Since image semantics are mainly encoded in the phase spectrum, this strategy transfers global appearance statistics such as illumination, contrast, and coarse texture from the target domain without explicitly altering class labels. Therefore, FDA can be used to synthesize target-style training images from labeled source samples and train a standard classifier in a fully supervised manner on the translated source set.

Given a source image x_s and a target image x_t , let their Fourier transforms be

$$\mathcal{F}(x) = A(x) \exp(jP(x)), \quad (10)$$

where $A(x)$ and $P(x)$ denote the amplitude and phase spectrum respectively. FDA first extracts the source amplitude A_s , target amplitude A_t , and source phase P_s , then replaces only the centered low-frequency band of A_s using a binary mask M_β parameterized by β . The mixed amplitude can be written as

$$\tilde{A}_s = (1 - M_\beta) \odot A_s + M_\beta \odot A_t, \quad (11)$$

and the translated source image is reconstructed by

$$\hat{x}_{s \rightarrow t} = \mathcal{F}^{-1} \left(\tilde{A}_s \exp(jP_s) \right). \quad (12)$$

In our implementation, M_β is a square low-frequency window centered in the shifted spectrum, where the side length is proportional to $\beta \cdot \min(H, W)$. A larger β injects more target-domain style, whereas a smaller β preserves more source-domain structure.

Following the implementation in our codebase, each labeled source image is randomly paired with one unlabeled target-training image, and FDA is applied offline to export translated versions of both ‘source_train’ and ‘source_val’. We then train the same ResNet-18 classifier as the source-only baseline on the FDA-translated source set, using the original source labels as supervision. The classification objective is

$$\mathcal{L}_{FDA}(f) = \mathbb{E}_{(x_s, y_s) \sim X_S} [\ell_{ce}(f(\hat{x}_{s \rightarrow t}), y_s)], \quad (13)$$

where f denotes the classifier and ℓ_{ce} is the cross-entropy loss. Model selection is performed on the FDA-translated validation split, and the final performance is reported on the real target-domain test set. This design makes FDA a lightweight UDA baseline that directly tests whether low-frequency style alignment alone is sufficient to improve target-domain generalization.

3.1.3 CycleGAN & Spectral CycleGAN

This UDA approach leverages the aforementioned standard CycleGAN and Spectral CycleGAN model. The model would generate a style-transferred dataset using labeled source images. Subsequently, a pre-trained ResNet-18 is fine-tuned on this synthesized dataset and finally evaluated on the target domain to measure its classification accuracy.

3.1.4 CyCADA

CyCADA [2] is a framework that integrates pixel-level domain adaptation via CycleGAN with feature-level adaptation based on the ADDA methodology [4]. By incorporating a semantic consistency loss (Equation (15)) into the CycleGAN module, CyCADA ensures that the transformed images maintain pixel-wise label alignment with the source domain for downstream tasks. Furthermore, the feature-level adversarial training (Equation (16)) leverages the ”pre-aligned” results from CycleGAN to further eliminate source-domain noise

within high-level semantic features, thereby enhancing the overall performance of Unsupervised Domain Adaptation (UDA).

The overall loss function of the proposed method can be expressed as Equation (14) to (16)

$$\begin{aligned}
L_{CyCADA}(f_T, X_S, X_T, Y_S, G_{S \rightarrow T}, G_{T \rightarrow S}, D_S, D_T) \\
= L_{task}(f_T, G_{S \rightarrow T}(X_S), Y_S) \\
+ L_{GAN}(G_{S \rightarrow T}, D_T, X_T, X_S) + L_{GAN}(G_{T \rightarrow S}, D_S, X_S, X_T) \\
+ L_{GAN}(f_T, D_{feat}, f_S(G_{S \rightarrow T}(X_S)), X_T) \\
+ L_{cyc}(G_{S \rightarrow T}, G_{T \rightarrow S}, X_S, X_T) + L_{sem}(G_{S \rightarrow T}, G_{T \rightarrow S}, X_S, X_T, f_S)
\end{aligned} \tag{14}$$

$$\begin{aligned}
L_{sem}(G_{S \rightarrow T}, G_{T \rightarrow S}, X_S, X_T, f_S) = L_{task}(f_S, G_{T \rightarrow S}(X_T), p(f_S, X_T)) \\
+ L_{task}(f_S, G_{S \rightarrow T}(X_S), p(f_S, X_S))
\end{aligned} \tag{15}$$

$$\begin{aligned}
L_{GAN}(f_T, D_{feat}, f_S, G_{S \rightarrow T}, X_S, X_T) = \mathbb{E}_{x_t \sim X_T} [\log D_{feat}(f_T(x_t))] \\
+ \mathbb{E}_{x_s \sim X_S} [\log(1 - D_{feat}(f_S(G_{S \rightarrow T}(x_s))))]
\end{aligned} \tag{16}$$

Where X_S (with labels Y_S) and X_T denote the source and target domain data respectively; $G_{S \rightarrow T}$ and $G_{T \rightarrow S}$ represent the cross-domain generators for image translation, while D_S and D_T are the corresponding pixel-level discriminators; f_S refers to the pre-trained model for source domain, whereas f_T is the target task model being optimized; D_{feat} denotes the feature-level discriminator responsible for domain alignment.

In this experiment, f_S adopts the same ResNet-18 architecture as the source-only model to guide the generation of the semantic consistency loss. Subsequently, we evaluate the performance of the CyCADA framework across three distinct configurations: utilizing only pixel-level adaptation, combining both pixel-level and feature-level adaptation, and integrating the results from our previously mentioned Spectral CycleGAN with CyCADA’s feature-level adaptation (where the style-transferred outputs from CyCADA are maintained during ADDA training).

3.2 Experimental Setup

We conducted image-to-image translation experiments using aforementioned models across four benchmarks: MNIST $\rightarrow$ USPS (Digits), Amazon $\rightarrow$ Webcam (Office-31), Art $\rightarrow$ Real-World (Office-Home), and Photo $\rightarrow$ Sketch (PACS).

In this experimental section, all ResNet-18 backbones are fine-tuned from Torchvision’s pre-trained models. The training process employs a cosine annealing scheduler with an initial learning rate of 0.0001, complemented by an early stopping strategy to secure the optimal model performance. The training configuration for the CycleGAN component remains consistent with the settings described in Task I.

3.3 Results and Analysis

The final experimental results are shown in Table 1.

From the experimental results, it can be observed that the performance of different UDA methods varies across datasets. On the MNIST $\rightarrow$ USPS benchmark, the full CyCADA framework achieves the best performance, with the results being largely consistent with those reported in the original paper. However, on the Amazon $\rightarrow$ Webcam dataset, the combination of Spectral CycleGAN and feature adaptation outperforms CyCADA, reaching the best accuracy. Notably, on the Art $\rightarrow$ Real-World dataset, all CycleGAN-based UDA methods lead to negative optimization compared to the baseline, with FDA yielding the most superior results. Finally, for the Photo $\rightarrow$ Sketch task, Spectral CycleGAN achieves the highest accuracy, whereas methods involving CyCADA method exhibit a significant performance degradation.

Table 1: Quantitative Comparison of UDA Approaches across Different Benchmarks

Method	MNIST→ USPS	amazon→ webcam	art→ real world	photo→ sketch
source-only	81.22%	48.43%	48.82%	25.94%
FDA	79.57%	52.75%	55.10%	31.81%
cycleGAN (standard)	90.93%	53.21%	46.91%	53.93%
cycleGAN (spectral)	95.76%	56.98%	46.57%	55.36%
CyCADA pixel only	96.11%	46.54%	45.83%	23.87%
CyCADA pixel + feat	96.16%	53.58%	45.31%	37.80%
cycleGAN (spectral) + feat	96.06%	59.50%	45.93%	44.34%

The aforementioned results underscore the inherent complexity of the domain gap in UDA tasks, demonstrating that no single model can universally excel across all transfer learning scenarios. While FDA achieves robust improvements on the latter three datasets, its reliance on simple low-frequency swapping tends to introduce artifacts in digit style transfer, thereby compromising classification accuracy. In contrast, CycleGAN performs well on the MNIST → USPS, Amazon → Webcam, and Photo → Sketch benchmarks, with the integration of spectral-based loss yielding positive refinements. Observations of the original datasets suggest that these three benchmarks exhibit distinct and consistent discrepancies in low-frequency features, such as color and illumination. However, the domain gap in the Art → Real-World dataset is primarily characterized by structural differences in object shapes. This poses a significant challenge for CycleGAN, which is designed to preserve the original content’s geometry. Furthermore, the inevitable blurriness and artifacts in the generated images lead to a classification performance that falls below the baseline.

The performance of CyCADA appears more complex than anticipated. While it achieves the expected state-of-the-art results on the digit dataset, its pixel-only adaptation surprisingly underperforms compared to the standard CycleGAN on other benchmarks. Regarding this phenomenon, we hypothesize that the ResNet-18 model, intended to guide semantic alignment during CycleGAN training, may have instead facilitated shortcut learning. Specifically, the generator might have learned to produce subtle pixel-level features designed to “deceive” the ResNet-18 classifier. Although these spurious features are imperceptible to the human eye, they lead to severe overfitting when the style-transferred images are subsequently used to train the task network, ultimately degrading the overall UDA performance.

4 Conclusion

In this project, we systematically studied image-level domain adaptation from both the perspective of visual translation quality and downstream classification performance. Through Task I, we observed that both standard CycleGAN and Spectral CycleGAN are capable of producing visually plausible cross-domain translations. However, visual inspection alone is insufficient to determine whether a translated image is truly beneficial for UDA, motivating the quantitative benchmarking conducted in Task II.

The downstream classification results demonstrate that no single UDA approach consistently dominates across all datasets. CyCADA achieves the strongest performance on MNIST → USPS, indicating that semantic consistency and feature-level alignment are particularly effective for digit adaptation. On Amazon → Webcam and Photo → Sketch, the Spectral CycleGAN-based approach delivers the best overall performance, suggesting that frequency-aware style regularization can improve the usefulness of translated training samples. In contrast, FDA performs best on Art → Real-World, showing that lightweight low-frequency spectrum replacement can still be competitive when the domain discrepancy is better characterized by global appearance statistics than by complex geometric transformation.

At the same time, our experiments also reveal important limitations of input-space UDA. FDA may fail when aggressive spectral replacement corrupts semantic content, while CycleGAN-based methods can suffer from blurry reconstructions, artifact accumulation, and even negative transfer when the true domain gap is dominated by structural differences. The mixed results of CyCADA on non-digit datasets further suggest

that stronger alignment objectives do not automatically guarantee better generalization, especially when the translated images themselves are imperfect or when the semantic guidance model introduces undesirable biases.

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